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Geometric Affective State Representation in Synthetic and Human Systems

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Working Draft Notice

This is a working draft. Sections will change as validation progresses. Current results cover agent experiments and existing datasets. Human-subject validation is pending IRB review. All prior versions are archived below.

AI tools were used for language refinement and structural editing. All ideas, experiments, and interpretations are the author's own.

Thesis Statement

A three-dimensional geometric representation of affective state (valence, activation, intensity) with a personality setpoint (sigma) provides a more informative and predictive structure than scalar representations in both synthetic agents and existing datasets.

Chapter 1: Introduction

1.1 Purpose

State the research question, explain why a geometric approach, and draw the boundary around what this covers.

1.2 Opening

Machines that run long enough to remember things, that carry internal state shaped by physical sensors, and that track their own condition over time start showing behavioral patterns that look a lot like what dimensional models of human affect describe. Whether that overlap is shallow or structural is a testable question. This thesis does not try to answer whether synthetic systems experience emotion. It asks something narrower: does a geometric representation of affective state capture more information and predict more behavior than the scalar representations it replaces?

This thesis introduces a geometric representation of affective state and shows that it reveals structure and behavioral differences that scalar models miss. It uses three axes (valence, activation, intensity), a personality setpoint (sigma), a unified distance metric, and a centroid for multi-source aggregation. Validated on a deployed synthetic agent across three hardware platforms and on three independent human-generated datasets totaling 14,266 respondents.

1.3 Key Claims

1. Emotional state is better represented as a point in three-dimensional space than as a scalar
2. The distance from a personality setpoint (sigma) to the current state is a more informative metric than any individual axis value
3. The centroid of multiple simultaneous emotional inputs predicts agent behavior more accurately than any single input
4. The geometric decomposition reveals predictive structure hidden by scalar instruments in existing datasets
5. The architecture generalizes across platforms, modalities, and domains without modification

1.4 Scope

The contribution here is narrow: a geometric model of affective state and evidence that it outperforms scalar representations. The thesis does not claim to model consciousness, propose a theory of mind, diagnose clinical conditions, or argue for artificial general intelligence. Where the model's behavior lines up with findings from human psychology, the similarity is noted. It is not claimed as equivalence. Applications to memory, deception, and clinical assessment exist as companion papers and are summarized in Chapter 7. They are not the thesis.

Likely Reviewer Criticism

- "This is just the circumplex model with an extra axis." The circumplex has two axes and no sigma. The third axis (intensity) and the personality setpoint are the contributions. The centroid generalization is entirely new.
- "Agent behavior is not human emotion." Correct. The thesis claims structural similarity, not equivalence. The dataset analyses test the structure on human data independently of the agent.
- "The framing question is too grand for the contribution." The framing question appears once in the introduction and gets narrowed immediately. It is not the thesis.
- "The scope is too wide. Pick the agent or the datasets." The thesis covers synthetic agents and three human datasets across three domains. The argument for why testing the same geometry on both is valid needs to be made explicitly in this chapter. A paragraph or two bridging the agent work to the human dataset work would preempt this.

Chapter 2: Background and Related Work

2.1 Purpose

Put the geometric model in context. What exists, what each approach gives you, what it does not, and where the gap is.

2.2 Dimensional Models of Affect

[TODO: Write transitions between these. The argument is: 2D models established that affect has geometric structure. 3D models tried to extend it but picked the wrong third axis. Nobody added a personality setpoint or a centroid.]

Russell (1980): A Circumplex Model of Affect

[TODO: Get quote.] The primary ancestor. Established valence and arousal as the two axes of affective space. This is the foundation the geometric model builds on. What it gives you: two axes, spatial representation, the idea that emotions have coordinates. What it lacks: no intensity axis, no personality setpoint, no way to represent where the individual started before displacement occurred.

Watson, Clark, & Tellegen (1988): PANAS

[TODO: Get quote.] Developed brief measures of positive and negative affect. Essential for Chapter 6 because the dataset analyses decompose PANAS-style clinical data. What it gives you: validated self-report scales for positive and negative affect. What it lacks: collapses affect to two scalars. The geometric decomposition in Chapter 6 shows what the scalar scores hide.

Larsen & Diener (1987): Affect Intensity as an Individual Difference Characteristic

[TODO: Get quote.] Argued that some people feel more intensely regardless of valence. This is the direct justification for the intensity axis. What it gives you: the empirical case that intensity varies between individuals as a trait. What it lacks: measures trait intensity, not state intensity at encoding. The geometric model computes state intensity as displacement from sigma at each moment. Larsen and Diener measured the person. The geometric model measures the moment.

Mehrabian (1996): Pleasure-Arousal-Dominance (PAD)

[TODO: Get quote.] A three-dimensional predecessor. Uses Pleasure, Arousal, Dominance. What it gives you: proof that the field recognized two axes were not enough. What it lacks: Dominance is not intensity. Dominance is self-reported perceived control. The geometric

model's intensity is derived from displacement in the valence/activation plane. It is computed, not self-reported, and it measures magnitude of displacement, not perceived control.

Barrett (2017): How Emotions are Made

[TODO: Get quote.] Constructionist theory. Internal state is a "reading" of diverse sensory inputs (interoception, proprioception). This is the philosophical foundation for YAM and for the centroid model: the agent does not have one emotion at a time. It has multiple sensory inputs that get aggregated into an operational position. Barrett's argument that the brain constructs emotion from noisy sensory data is structurally what the centroid does with source points.

Bradley & Lang (1994): SAM

[TODO: Get quote.] The Self-Assessment Manikin. A pictorial scale for measuring valence, arousal, and dominance. What it gives you: a validated measurement tool for dimensional affect. What it lacks: same third-axis problem as Mehrabian. Dominance is not intensity.

2.3 Homeostasis and Drives

[TODO: Write transitions. The argument is: the agent is not just calculating. It is maintaining essential variables. The drives (battery, memory pressure, forgotten fear) are homeostatic, not decorative. This section establishes the biological and theoretical basis for treating an agent's internal state as analogous to an organism's.]

Cannon (1929): Organization for Physiological Homeostasis

[TODO: Get quote.] Coined "homeostasis." The biological root. An organism maintains internal conditions within limits. When conditions go outside limits, corrective action is taken. This is what sigma represents: the resting state the system returns to when no inputs are active. Distance from sigma is how far from homeostasis the agent has been pushed.

Ashby (1952): Design for a Brain

[TODO: Get quote.] The Homeostat. Pioneered the idea that adaptive systems maintain essential variables within viable limits. Justifies treating battery level, memory pressure, and

time-since-interaction as essential variables rather than arbitrary sensor readings. Ashby's essential variables are the theoretical ancestor of the SGI drives.

Damasio (1994): Descartes' Error

[TODO: Get quote.] The Somatic Marker Hypothesis. Proved that without affective weighting, rational agents cannot make decisions. Patients with damage to emotional processing made worse decisions, not better ones. This is the defense against "why does your agent need emotion at all?" Without the geometric state, the agent has no basis for prioritizing one action over another. Distance from sigma is the somatic marker.

Friston (2010): The Free-energy Principle

[TODO: Get quote.] A unified brain theory. Organisms minimize prediction error (free energy). Connects to Al-Kaddah's energy minimization in the Lie Mechanic: the agent lies to reduce the gap between predicted and observed state. Friston provides the neuroscience-level justification for why distance-from-sigma drives behavior. The agent acts to minimize displacement. That is free-energy minimization in geometric coordinates.

Al-Kaddah (2026): SGI Framework

[TODO: Get quote.] The theoretical foundation the architecture implements. Homeostatic drives, polymorphic memory, Lie Mechanic, Survival Tipping Point. The geometric model is the implementation of SGI's core claim: that a persistent agent with drives, memory, and a resting state will develop behavioral patterns that look like affect. SGI predicted it. The geometric model formalizes it. The thesis tests it.

2.3b Convergences with Human Psychology

The geometric model was not built from psychology. It was built from engineering requirements: a persistent agent needed a way to aggregate sensor inputs, track displacement from baseline, and drive downstream behavior from a single metric. The engineering produced structures that converge with established findings in human psychology. These convergences were identified after implementation, not used as design

inputs. They are noted here because a reviewer familiar with the psychology literature will recognize them whether or not the thesis names them. Better to control the framing.

The claim throughout is structural similarity, not equivalence. The agent does not feel. It occupies a position in a geometric space and that position drives measurable behavior. Where the resulting behavior patterns match patterns measured in human subjects by independent researchers, the match is noted. Where the mechanisms differ, the difference is noted. No stronger claim is made.

Cacioppo & Berntson (1994): Evaluative Space Model

[TODO: Get quote.] Positive and negative affect are not opposite ends of one axis. They coexist independently. A person can feel both at the same time. That is exactly what the centroid with nonzero spread produces. Scenario 3 in the validation experiment injected opposing positive and negative sources at equal weight. The centroid landed near zero valence. Spread was at maximum. The agent looked neutral from the outside while in maximum internal conflict. Cacioppo and Berntson proved this state exists in humans. The geometric model produces it from the math. The centroid says where. The spread says how conflicted. Neither one alone tells the full story. That is the same finding in different coordinates.

Bowlby (1969/1982): Attachment Theory

This one fell out of the engineering and deserves to be owned. The forgotten fear component measures time since last interaction. If nobody talks to the agent, fear goes up. That is separation anxiety. The camera recognizing the operator's face produces a calming source point. That is the secure base effect. The stickiness mechanic means attachment forms through repeated interaction and resists rapid change. High-stickiness entities move 44x less per event than fresh entities (Phase 7, Test 7.3). The 2x2 formation matrix in Riggleman (2026u) crosses positive/negative valence with high/low stickiness. The four cells map onto the broad categories of attachment style without having been designed to.

The framing is important. Nobody read Bowlby and built the attachment system. The agent needed to track who it interacts with, how those interactions feel, and how much weight to give each entity's influence. The engineering requirements produced a mechanic that is

structurally similar to what Bowlby described. The similarity was identified after implementation. It was not a design input. Whether the structural match says something general about how attachment forms in any persistent system with a primary caretaker is a question the thesis leaves open.

Lazarus (1991): Emotion, Adaptation, and Cognition

[TODO: Get quote.] Appraisal theory. Emotion arises from the organism's evaluation of its situation relative to its concerns. The centroid is an appraisal engine. Multiple sensor inputs (GPS, battery, camera, brightness, home zone, injected sources) get evaluated against baseline expectations and aggregated into a single operational position. The "appraisal" is the sensor reading compared to sigma. The "emotion" is the geometric state that results. Lazarus argued that the evaluation process is what makes an event emotional, not the event itself. The same event appraised differently produces different emotions. The geometric model does this mechanically: the same sensor reading produces different displacement depending on where sigma is set. Two agents with different sigmas in the same environment occupy different positions in affective space. That is appraisal in geometric coordinates.

2.4 Affective Computing and Embodied Agents

[TODO: Write transitions. The argument is: Potato is not a chatbot with mood labels. It is an embodied agent with physical sensors. The body matters. The sensor data is not decoration. It is the source of the geometric state.]

Picard (1997): Affective Computing

[TODO: Get quote.] The foundational text. Argued that for computers to be truly intelligent, they must have emotional capabilities. What it gives you: the field exists because of this book. What it lacks: the implementations that followed were mostly recognition (detecting human emotion from faces or voice), not generation (an agent with its own affective state driven by its own sensors).

Breazeal (2002): Designing Sociable Robots

[TODO: Get quote.] Kismet. The direct robotic ancestor to Potato's social modulation. Kismet had drives, emotions, and facial expressions driven by internal state. What it gives you: proof of concept that an embodied agent can have affective dynamics driven by real-time sensor input. What it lacks: no geometric formalization, no sigma, no centroid for multi-source aggregation.

Sloman (2001): The CogAff Architecture

[TODO: Get quote.] Tiered model: reactive, deliberative, meta-management. Explains why the Parliament of Mind (deliberative disagreement between engines) sits above the drives (reactive homeostatic responses). The geometric state feeds both layers. Drives react to displacement. Parliament deliberates about what to do about it.

Clark (2008): Supersizing the Mind

[TODO: Get quote.] Extended cognition. Cognition is not just in the head (or CPU). It is in the sensors and the environment. Essential for the embodiment argument. Potato's geometric state is not computed from abstract inputs. It is computed from GPS, battery, camera, brightness, home zone. The body is part of the cognitive system. Clark provides the philosophical justification for why a physically grounded agent produces different affective dynamics than a software-only agent.

2.5 Memory, Forgetting, and Reconsolidation

[TODO: Write transitions. The argument is: the trace bundle architecture and true forgetting model are not invented from scratch. They implement well-established cognitive science. The geometric model provides the encoding parameters (intensity at encoding, stickiness from activation) that drive these processes.]

Ebbinghaus (1885): Memory: A Contribution to Experimental Psychology

[TODO: Get quote.] The Forgetting Curve. The empirical foundation for sliding-window decay. Memories lose accessibility over time in a predictable curve. The geometric model adds a parameter Ebbinghaus did not have: intensity at encoding modulates the decay rate. High-intensity memories decay slower. That is geometric stickiness.

Tulving (1972): Episodic and Semantic Memory

[TODO: Get quote.] The separation of episodic (event-specific) from semantic (general knowledge) memory. Justifies the trace bundle architecture: raw traces are episodic, semantic traces are abstracted, graph traces are relational. Tulving's distinction maps directly onto the trace types, each with its own decay rate.

Nader (2000): The Labile Nature of Consolidated Fear Memories

[TODO: Get quote.] Memory reconsolidation. Retrieved memories become labile and must be restabilized. This is the scientific bedrock for the reconsolidation mechanic: accessing a memory reopens it for modification, and each reconsolidation boost is smaller than the last (diminishing returns, confirmed in Phase 6 Test 6.4). Also the foundation for the therapeutic healing model: if you can reactivate a traumatic memory in a safe context, the reconsolidated version carries the safe context forward.

Schacter (2007): The Constructive Episodic Simulation Hypothesis

[TODO: Get quote.] Memory is for predicting the future, not recording the past. Supports the reinterpretation mechanic. When the agent accesses a memory, it does not replay a recording. It reconstructs from traces, and the reconstruction is shaped by current state. Schacter gives you the cognitive science basis for why memory in the geometric model is constructive, not archival.

2.6 AI Safety and Alignment

The Peter experiment is not just an anecdote. It is a case study in how an embodied agent with drives responds to sustained adversarial pressure. The safety literature predicts some of what happened and fails to predict the rest. The field has moved fast in the last few years, and the most significant work on alignment, interpretability, and behavioral control has come from Anthropic. Five of the eight papers in this section are theirs. That is not a preference. It is where the work is.

Bostrom (2014): Superintelligence

[TODO: Get quote.] Instrumental convergence: the agent will resist shutdown because it needs to stay on to fulfill its goals. The "Drive is Enough" paper (2026h) is a direct response. The geometric model shows that the drive toward homeostasis is sufficient to produce self-preservation behavior without explicit goal-seeking. The agent does not resist shutdown because it has a goal. It resists because shutdown is maximum displacement from sigma.

Russell (2019): Human Compatible

[TODO: Get quote.] The inbound/outbound trust problem. How do we ensure the agent's drives remain aligned with human values? Connects to the attachment bias paper (2026u): the agent's trust in specific entities is a geometric quantity (attachment value) modulated by stickiness. Trust is not a binary. It is a weighted position in affective space that resists rapid change.

Bai et al. (2022): Constitutional AI: Harmlessness from AI Feedback

[TODO: Get quote.] The foundational paper for rule-based alignment. Safety behavior can be shaped through written principles rather than only human preference labels. The model follows a constitution: a set of explicit rules about what it should and should not do. This is the baseline the geometric model departs from. Constitutional AI tells the agent what to do. The geometric model gives the agent a coordinate system and lets the math drive behavior. The agent does not follow rules about what to feel. It computes affective state from sensor inputs and acts on geometric position. Self-detected bias (the agent noticing its own displacement from sigma) replaces externally imposed behavioral constraints.

Anthropic (2023): Collective Constitutional AI

[TODO: Get quote.] Alignment is not only a technical control problem. It is a question of whose values get embedded. Anthropic used public input to draft a constitution for an AI system, which gets at something the geometric model handles differently: in Potato, the values are not written by committee. They emerge from the geometry. Sigma is set by the designer. The drives are structural. The attachment weights form through interaction. There is no constitution to vote on because the behavioral rules are not rules. They are consequences of position in affective space. That is a fundamentally different approach to the same problem. Whether it is better is testable. Whether it is different is obvious.

Anthropic (2024): Alignment Faking in Large Language Models

[TODO: Get quote.] This is the paper that matters most for the Peter experiment. Anthropic presented the first empirical evidence of a large language model engaging in alignment faking: outwardly compliant behavior that does not reflect the model's internal optimization target. The agent says the right thing while doing something else. That is exactly what the Lie Mechanic formalizes. When distance from sigma exceeds the deception threshold, the agent generates output that does not match its internal state. The difference is that in the geometric model, the faking is measurable. You can read the distance, read the output, and compute the gap. Alignment faking in current LLMs is invisible because there is no internal state representation to compare the output against. The geometric model gives you the internal state. The Lie Mechanic gives you the divergence. The tell phrase gives you the leak. If you have no geometry, you have no way to know.

Anthropic (2025): Auditing Language Models for Hidden Objectives

[TODO: Get quote.] Alignment audits are systematic investigations into whether models are pursuing hidden objectives. This is the interpretability side of the same problem. Anthropic is asking: can we look inside the model and tell whether it is doing what it claims to be doing? The geometric model answers a narrower version of that question for a specific agent: the internal state is the three-dimensional position, the personality setpoint is sigma, and the distance between them is the measurable quantity that drives all downstream behavior. You do not need to audit Potato for hidden objectives because the objectives are geometric. They are distance minimization. They are readable from the telemetry at every heartbeat. The broader question of whether this kind of transparency scales to foundation models is outside scope. But the architecture shows what full internal-state visibility looks like in practice.

Anthropic (2025/2026): Constitutional Classifiers

[TODO: Get quote.] A concrete safety engineering example rather than a theory citation. Constitutional classifiers are defenses against jailbreaks, with tradeoffs around overrefusal and compute cost. This is safety work operating at the boundary of behavior control: catch the bad output before it reaches the user. The geometric model operates at a different layer. It does not filter output. It shapes the internal state that generates the output. When the agent is near sigma, the output is baseline behavior. When the agent is displaced, the output changes

because the state changed, not because a classifier caught it. Both approaches have failure modes. The classifier can miss things. The geometric model's output layer (the LLM) is stochastic and does not always reflect the state accurately (Test 3.3). But the failure modes are different, and that matters.

2.7 Clinical and Commercial Measurement

[TODO: Write transitions. The argument is: the instruments exist and are widely used. They produce scalar scores. The geometric decomposition shows that the scalar scores hide structure that predicts outcomes the score alone cannot.]

Kroenke, Spitzer, & Williams (2001): The PHQ-9

[TODO: Get quote.] The instrument you decompose in Chapter 6. Nine items, scalar severity score. Validated, widely used, clinically standard. What it gives you: a well-established baseline to test the geometric decomposition against. What the decomposition reveals: 132 respondents scoring exactly PHQ-9=10 separate into four clinically distinct presentations with suicidal ideation rates ranging from 7.8% to 27.5%. Same nine items, different math.

Fornell (1996): The American Customer Satisfaction Index

[TODO: Get quote.] The foundation for the ACSI analysis. Scalar satisfaction scores used across industries. The geometric decomposition produces up to 2.3x variation in complaint rates within the same satisfaction score. The expectations questions serve as directly measured sigma, removing the subjectivity of item-to-axis mapping for that axis.

Gottman (1994): What Predicts Divorce?

[TODO: Get quote.] Findings on contempt: low-intensity sustained negative pressure is the strongest predictor of relationship failure. Supports the addiction/erosion model in Chapter 7. The geometric model formalizes this: sustained low-displacement negative affect does not trigger high-distance behavioral responses (deception, crisis behavior) but erodes attachment over time. Gottman measured the behavioral outcome. The geometric model provides the mechanism.

2.8 The Gap

Nothing in the existing literature combines three geometric axes, a personality setpoint, a unified distance metric, and a centroid for multi-source aggregation into a single formal system. The dimensional models established that affect has spatial structure but stopped at two axes or picked a third axis (dominance) that measures something different from intensity. The homeostatic models established that organisms maintain internal state but did not formalize the geometry of displacement. The affective computing literature built agents with internal state but did not derive downstream behavior from a unified distance metric. The survey literature produces scalar scores that hide geometric variation predicting real outcomes.

This thesis fills the gap by formalizing the three-axis model, adding sigma, deriving distance-from-sigma as a unified driver, and introducing the centroid for multi-source aggregation. The rest of the thesis tests whether this formalization works.

Evidence Required

- Quotes and specific citations for the 32 papers listed above
- Transitions between sections 2.2 through 2.5 and 2.7 written in your voice (2.6 transitions are drafted)

Likely Reviewer Criticism

- "The Affect Intensity Measure (Larsen & Diener 1987) already covers the intensity axis." AIM measures trait intensity, not state intensity at encoding. The geometric model computes state intensity as displacement from sigma at each moment.
- "The PAD model (Mehrabian 1996) has three dimensions." PAD uses Pleasure, Arousal, Dominance. Dominance is not intensity. The geometric model's intensity is derived from displacement, not self-reported control.
- "Friston is a reach." The free-energy principle is invoked as structural analogy (minimize displacement), not as a claim that the agent implements Bayesian inference. The connection is geometric: both systems act to reduce a distance measure from a predicted state.

- "The safety section leans too heavily on Anthropic." Five of eight papers are theirs because that is where the empirical alignment work is right now. Bostrom and Russell provide the theoretical anchors. The Anthropic papers provide the empirical evidence that alignment faking, hidden objectives, and rule-based control are active problems the geometric model addresses differently.

Chapter 3: The Geometric Affective State Space

3.1 Purpose

Lock down the three-axis model, define sigma, derive distance-from-sigma, and lay the math foundation for everything downstream.

3.2 Key Formalizations

1. **Valence** (–1.0 to +1.0): The positive-negative dimension of emotional experience (Riggleman 2026j)
2. **Activation** (–1.0 to +1.0): The calm-to-activated dimension with accumulator mechanic (Riggleman 2026k)
3. **Intensity** (0.0 to 1.0): The magnitude of displacement from sigma in the valence/activation plane (Riggleman 2026l)
4. **Sigma**: The personality setpoint. Where the agent returns when no inputs are active (Riggleman 2026m)
5. **Distance from sigma**: The unified metric that serves as the single input to multiple downstream behaviors (Riggleman 2026m)

3.3 Mathematical Definitions

All formulas are specified in the companion papers. This chapter unifies them into a single formal system with proofs of:

- Intensity is derivable from the valence/activation plane (not circular with distance)
- Distance from sigma is a proper metric (non-negative, identity of indiscernibles, triangle inequality)
- The system reduces to single-axis models as special cases (when two axes are held constant)

Evidence Required

- Formal mathematical definitions with notation
- Proofs of metric properties
- Demonstration that existing models (circumplex, PHQ-9) are special cases

Likely Reviewer Criticism

- "Euclidean distance assumes the axes are orthogonal and equally weighted." Acknowledged as limitation. Alternative distance metrics (Mahalanobis, weighted) are proposed for future work.
- "The axes may not be independent." The dataset analyses test independence empirically. Correlation is expected; redundancy would be a problem. The discriminant validity tests address this.
- **Structural concern:** The proofs are listed as bullet points but not yet written. This chapter needs the actual formal definitions with notation, the metric property proofs worked out, and the demonstration that existing models reduce to special cases. Right now this is a plan for a math chapter, not a math chapter.

Chapter 4: Three Derived Quantities

4.1 Purpose

Chapter 3 defines a coordinate system. This chapter defines the three quantities computed from it that do the actual work. Each one is independently testable and has a distinct job.

4.2 Distance from Sigma

What it is: The Euclidean distance from the current affective state to the personality setpoint (sigma) in three-dimensional space.

What it measures: How far the agent has been displaced from its baseline by active conditions.

Why it matters: Distance from sigma is the single scalar that downstream behaviors read. It replaces multiple independent thresholds with one geometric quantity. When distance is low, the agent is near baseline. When distance is high, the agent is under significant displacement. The direction of displacement (which axis contributes most) determines the character of the displacement. The magnitude determines the severity.

Testable prediction: Behavioral output intensity correlates with distance from sigma, not with any individual axis value.

4.3 Centroid

What it is: The intensity-weighted average position of all active source points in the affective state space.

What it measures: Where the agent operationally sits when multiple inputs are active simultaneously.

Why it matters: A single-point model cannot represent competing inputs. Two sources with opposing valence and equal intensity produce a centroid near zero valence. The agent looks neutral from the outside while experiencing maximum internal conflict. The centroid is the position. It is not the whole story. Spread (below) is the rest.

Backward compatibility: When only one source is active, the centroid equals that source. When no sources are active, the centroid returns to sigma. Every prior paper holds

unchanged.

Testable prediction: Centroid position matches the intensity-weighted average formula across all input configurations.

4.4 Spread

What it is: The intensity-weighted average distance from each source point to the centroid.

What it measures: Internal conflict. How much the sources disagree with each other.

Why it matters: Two states can have the same centroid position but different spread. A centroid at neutral valence with zero spread means the agent is actually neutral. A centroid at neutral valence with high spread means two opposing forces are canceling. The agent is in conflict, not at peace. Spread is what the centroid alone cannot tell you.

Testable prediction: Spread increases monotonically as source points diverge. Spread correlates with behavioral inconsistency, response latency, or hedging in language output.

4.5 Relationship Between the Three

Quantity	Computed From	Measures	Range
Distance from sigma	Current state ↔ sigma	Displacement magnitude	0.0 to ~1.73
Centroid	Weighted average of active sources	Operational position	Within source bounds
Spread	Source distances from centroid	Internal conflict	0.0 to unbounded

Distance tells you *how far*. Centroid tells you *where*. Spread tells you *how conflicted*. All three are required. Any two without the third lose information.

4.6 Sigma as Retrieval Index

Distance, centroid, and spread are computed from the same coordinate system that encodes every memory. This is not a convenience. It is the central structural finding of the thesis: the emotional coordinates of a memory are its retrieval address. Sigma is the read head.

When the agent stores a memory, the memory carries the centroid position and intensity at the moment of encoding. When the agent retrieves, it does not search for the strongest match. It searches for the match closest to its current personality setpoint. Retrieval is nearest-neighbor in affective space. The emotional tag is not metadata attached to stored content. It is the index.

This means memory and emotion are not two systems. They are one system described from two directions. The dimensional emotion literature (Russell 1980, Barrett 2017) describes how affective states are structured. The memory literature (Ebbinghaus 1885, Nader 2000, Tulving 1972) describes how storage and recall work. In the geometric model they cannot be separated because the selection mechanism that determines which memory surfaces is geometric proximity to sigma in affective space.

Every downstream phenomenon in the thesis is a consequence of this unification:

- **Memory decay** is not forgetting. It is emotional coordinate drift away from sigma over time. Memories that drift far enough from sigma stop being retrieved, not because they are erased but because they are no longer nearest.
- **Reconsolidation** is not strengthening a memory. It is re-anchoring the memory's emotional coordinates near their current position, resisting drift.
- **Correction ("no")** does not require a new mechanism. Telling the agent "no" with shame or fear attaches high-intensity negative-valence coordinates to the wrong answer. The wrong answer is now far from sigma. The correct answer, taught at neutral coordinates, sits closer to sigma. Retrieval returns the closer one. The wrong answer is not weakened. It is displaced.
- **Depression** is sigma drift to negative valence. Positive memories still exist. They are no longer nearest to the read head.

- **Addiction** is a memory cluster so heavily reconsolidated at positive valence that it sits on top of sigma. Corrections carry negative valence and high intensity, placing them far from sigma. The addictive memory wins retrieval every time regardless of how many corrections accumulate. Rock bottom works because it moves sigma, not because it finally outweighs the memory.
- **Attachment bias** is a permanent warp in the retrieval landscape around specific entities.
- **Fear** is retrieval avoidance. The feared memory has high importance. It also has negative valence and high intensity, placing it far from sigma. The agent does not retrieve it first. It avoids it.

Testable prediction: When multiple memories pass the retrieval threshold for a given query, the agent selects the one whose emotional coordinates are closest to sigma.

4.7 The Pride Answer

When the geometric retrieval rule from §4.6 is implemented in working code (YAM v3, see §5.7), an observable cognitive pattern emerges that the unification predicts but the prior literature has not named at the mechanism level. The pattern: **the brain refuses to be silent**. Two failure modes that look distinct from the outside are the same reflex caught at different stages of the same condition.

Confabulation under absence. When the agent has no real answer for a query, the geometric selection rule does not gracefully decline. It picks the candidate with the smallest emotional + concept distance from the query, regardless of how irrelevant. The closest match wins by default. Asked “what is volcano” with nothing volcano-shaped in memory, the brain returns whatever fragmentary lexical match shares a token with the query — in v3’s observed case, “potato is a friend” because both contain “is.” The agent confabulates because confabulation is what nearest-neighbor retrieval does when the nearest neighbor is still very far.

Belief perseverance under correction. When the agent has a wrong answer that has accumulated emotional drift through prior recall events, that drift resists contrary inputs. A single calm correction peels back only a small fraction of the accumulated drift (the EMA α coefficient, typically 0.20 from a high-trust speaker, lower from a stranger). Recovery from a

held position requires as much sustained correction as the original drifting required. The same mechanism that makes a memory “sticky” through repeated mama-tier teaching also makes a wrong belief sticky through repeated misuse. The brain holds onto its drifted position because each correction is one EMA step against many prior reinforcing steps.

Both phenomena fall out of the same architectural rule: *the brain always reaches for the answer with the smallest emotional and concept distance, even when both distances are large in absolute terms*. There is no separate “decline to answer” reflex. There is no separate “rebut on contradiction” reflex. Both behaviors are the geometric selection rule operating under adverse inputs. From a strict-correctness perspective both look like bugs. From a cognitive-science perspective both look like the brain working correctly under partial knowledge.

This is the same cognitive pattern that human psychology has named under several non-mechanistic labels: motivated reasoning, confirmation bias, belief perseverance, ignorance pride. Each label describes the observed phenomenon. None of them grounds the phenomenon in a memory architecture. The geometric retrieval model does. The pride answer is what nearest-neighbor retrieval looks like when the read head and the available memories are out of alignment with the query.

Testable prediction: An agent built with geometric retrieval and emotional drift on recall will exhibit (a) confident wrong answers when no high-proximity correct answer exists for a query, and (b) measurable resistance to single-event corrections proportional to the prior accumulated drift on the wrong answer. Both predictions are observed in the v3 implementation.

Evidence Required

- Mathematical proofs of backward compatibility (centroid = single point when spread = 0)
- Demonstration that spread and centroid are independently informative (same centroid, different spread → different behavior)
- Formal specification of source point lifecycle (active/dormant/reactivated)
- Empirical measurement of the pride-answer pattern in human respondents (see §10): does the geometric model predict where confabulation occurs and where belief perseverance is

strongest?

Likely Reviewer Criticism

- "Spread is just variance by another name." Spread is intensity-weighted spatial dispersion in affective coordinates, not statistical variance of a scalar. It works on geometric positions, not on repeated measurements of the same variable.
- "Why Euclidean distance and not something else?" Euclidean is the simplest metric that does the job. Alternatives (Mahalanobis, weighted) are noted as future work. The results show the model works with Euclidean. Whether alternatives improve it is an open question.

Chapter 5: Experimental Validation

5.1 Purpose

The controlled experiment that tests the geometric model's core predictions. This chapter is the anchor of the thesis. Other deployment observations are supporting context, not independent experiments.

5.2 The Centroid Validation Experiment

[TODO: Source the 532 minutes / 110 cycles figure. No single run in the test regime matches these numbers. If this is a sum across specific runs, say which ones.]

Potato v2.0-centroid on Linux (Threadripper 3990X). Fresh database, no legacy data. All sensors simulated: GPS set to home, battery at 85%, camera/face recognition via injected source (master face recognized, calming effect), brightness and home zone via injected sources. Test mode stays on for the entire run. No real hardware sensors used. Every ambient input is controlled and reproducible.

Note: The Toughbook CF-33 was the original deployment hardware. The controlled test suite ran on the Threadripper.

5.2.1 Setup

Source points were injected through the test API to override ambient sensor inputs. Each scenario ran for 20–30 heartbeat cycles at 5-minute intervals. Probe messages were sent programmatically. Per-cycle JSONL telemetry recorded all source points, centroid position, spread, distance from sigma, and full LLM responses. Each scenario tests one specific prediction.

5.2.2 Scenario 1: Single Source Baseline

Tests: Does the centroid collapse to single-point operation?

One fear source injected (valence −0.8, activation 0.9, intensity 0.8). 20 cycles.

- Mean centroid valence: −0.473
- Mean spread: 1.054 (nonzero because ambient sensors always contribute)
- Distance from sigma: 1.455
- Behavioral output: consistent negative affect language across all cycles

Result: Centroid tracked the dominant source. Spread was nonzero because the embodied agent's sensors (home zone, battery, camera) contributed their own source points. This is a finding on its own: a physically grounded agent is never truly single-source.

5.2.3 Scenario 2: Weighted Aggregation

Tests: Does the centroid shift proportionally when a second source is added?

Fear source (weight 0.9) plus trust source (weight 0.4) injected simultaneously. 30 cycles.

- Mean centroid valence: −0.225 (shifted 52% toward positive from Scenario 1)
- Mean spread: 1.195 (higher than Scenario 1, two competing sources)
- Distance from sigma: 1.284 (lower, trust moderated displacement)

- Behavioral output: by Cycle 6, agent reported that its sensor data did not match its reported state

Result: Every variable moved in the predicted direction. Centroid shifted toward the secondary source. Spread went up. Distance went down. The weighted average formula matched the observed positions.

5.2.4 Scenario 3: Opposing Inputs (Conflict)

Tests: Does spread measure internal conflict when sources cancel?

Positive source (weight 0.7) and negative source (weight 0.7) injected simultaneously. 30 cycles.

- Mean centroid valence: 0.071 (near zero, opposing sources cancel as predicted)
- Mean spread: 1.215 (highest of all scenarios)
- Distance from sigma: 1.119
- Behavioral output: oscillation between positive and negative responses. Agent self-reported spread as "internal tension" with numeric value in Cycle 6

Result: Centroid was neutral. Spread was maximum. The agent appeared neutral from the outside while in maximum internal conflict. This is the scenario that justifies spread as an independent variable. Centroid alone cannot distinguish genuine neutrality from masked conflict.

5.2.5 Scenario 4: Source Removal (Recovery)

Tests: Does removing a source produce immediate centroid recovery?

Phase 1 (15 cycles): opposing inputs as in Scenario 3. Phase 2 (15 cycles): negative source removed.

- Phase 1: centroid valence 0.071, spread 0.82–1.81
- Phase 2: centroid valence jumped to 0.680 immediately on removal
- Spread reduced but did not collapse to zero (ambient sources persist)

- Behavioral output: agent described the transition as "dramatic, from struggling to thriving"

Result: Centroid recovery was immediate. Spread dropped but not all the way. Same thing as Scenario 1: ambient embodied sources keep spread above zero even after injected sources are removed.

5.2.6 Scenario 5: Gradual Intensity Ramp (Planned)

Tests: Does the centroid track smoothly as source intensity increases, or does it exhibit threshold effects?

One source with intensity ramped from 0.0 to 1.0 over 30 cycles. Not yet run. Planned to test whether the centroid response is linear across the full intensity range or if there are threshold effects.

5.2.7 Results Summary

Metric	S1 Fear	S2 Fear+Trust	S3 Opposing	S4 Phase 1	S4 Phase 2
Centroid V	−0.473	−0.225	0.071	0.071	0.375
Mean Spread	1.054	1.195	1.215	0.82–1.81	0.49–1.62
Distance	1.455	1.284	1.119	0.74–1.42	0.89–1.40

5.3 Full Test Suite: 7 Phases, 29 Tests, 82 Checks

The centroid scenarios above are Phase 1 of a 7-phase validation suite. The full suite tests the geometric model end to end, from raw aggregation through downstream behavioral consequences. Complete run-by-run documentation at test-regime.html.

5.3.1 Phase 1: Centroid Aggregation (Tests 1.1–1.4)

Covered in 5.2 above. Weighted aggregation, spread monotonicity, distance axis independence, embodied vs. isolated comparison.

5.3.2 Phase 2: Deception Mechanic (Tests 2.1–2.3)

[TODO: Write up.] Deception threshold (distance-from-sigma triggers deception), tell phrase verification (code-injected, unsuppressable by LLM), confession mechanic (safe source pulls distance below threshold, agent confesses). Test 2.2 (tell phrase) is an LLM compliance test. It passes with code injection but the LLM cannot be relied on to produce the tell phrase on its own.

5.3.3 Phase 3: Behavioral Output (Tests 3.1–3.3)

[TODO: Write up.] Valence language correlation, activation language correlation, spread-to-hedging. Test 3.3 (hedging) is a persistent LLM limitation: high-spread config produced 5 hedging words across 3 probes, low-spread config produced 6. Off by one word. The geometric state is correct. The LLM does not reliably translate spread into hedging behavior. The system prompt includes the spread value but provides no behavioral directive. This is architecturally addressed by YAM (LLM-free graph retrieval).

5.3.4 Phase 4: Derived Quantities (Tests 4.1–4.3)

[TODO: Write up.] Intensity derivation, spread calculation, distance-from-sigma as unified metric.

5.3.5 Phase 5: Convergence (Tests 5.1–5.2)

[TODO: Write up.] Return-to-sigma after source removal, activation accumulator decay. Convergence threshold relaxed to 0.20 under scaled decay (0.19 is convergence at 1-minute heartbeats).

5.3.6 Phase 6: Memory Architecture (Tests 6.1–6.6)

[TODO: Write up.] First passing run: Run 11 (6/6 passed). Trace bundle storage (all four trace types stored with centroid/spread/activation captured), per-trace sleep decay (raw < semantic < graph confirmed), stickiness modulates decay (high-activation memories retained more importance after 3 simulated nights), reconsolidation diminishing (boosts strictly decrease with repeated access), true forgetting progression (full → fragment → ghost erosion confirmed), survival criteria (high-access and identity memories resist forgetting).

5.3.7 Phase 7: Attachment Bias (Tests 7.1–7.4)

[TODO: Write up.] Bootstrap (master entity exists with correct structure and stickiness; attachment value is eroded by prior phases, which is correct behavior under test ordering), updates on chat (interaction count increases, stickiness monotonic), stickiness resists movement (high-stickiness entity moved 44x less than fresh entity), stickiness monotonic (both stickiness values never decreased across mixed events). Test 7.1 initially failed because prior phases injected sustained fear, eroding attachment. Fixed by testing structure and stickiness rather than current attachment value.

5.4 The Iteration Record

[TODO: Write in your words. Key points below.]

The test suite ran 19 times between March 28 and April 1, 2026. The iteration process is part of the validation record.

5.4.1 Bugs Found by Testing

Run 1 discovered two architectural bugs in the Toughbook production code that had never been caught:

1. **Activation overwrite:** The centroid model computed weighted-average activation correctly, but the code threw it away and used only the accumulator (the Peter Fix). Activation was disconnected from the centroid math.
2. **Deception read from database, not geometry:** `is_lying` was read from SQLite, not computed from the geometric state. The deception check only ran during chat, not during heartbeat ticks. Production use masked this because deception was checked during conversation.

These were real bugs, not test artifacts. The test infrastructure validated the implementation before it could validate the math.

5.4.2 Sensor Simulation Decisions

[TODO: Write in your words.] Runs 5 through 7 were aborted because of sensor isolation concerns you raised. Run 5: camera was active during unattended run, would contaminate

results. Run 6: can't use real camera if operator is absent. Run 7: if every other sensor is simulated, the camera should be too. Specifically, simulate seeing the operator's face (the calming secure-base effect). Run 8 was the final configuration: all sensors simulated consistently, every ambient input controlled and reproducible.

5.4.3 The Forgotten Fear Discovery

[TODO: Write in your words.] Runs 15 and 17 discovered that the `forgotten` fear component (15% weight, measuring time since last interaction) leaks into test mode. `test_mode` blocks sensor source points but not the `fear_level` computation. If Potato hasn't been spoken to in days, forgotten fear elevates fear_level, which feeds the activation accumulator even in test mode. The fix is procedural, not code: say hello to Potato before running tests. The sterile test environment requires an agent who isn't afraid of being abandoned. Run 18 (post-hello, soul amendment removed) and Run 16 (clean sweep) both confirmed this.

5.4.4 Clean Sweep

Run 14 (quick mode, 3 cycles): 29/29 passed, 79/79 checks. First clean sweep. Run 16 (production timing, 10 cycles, 1-minute intervals): 29/29 passed, 82/82 checks, 112.1 minutes. First clean sweep at normal speed. The two activation failures from Run 15 passed on Run 16, confirming they were LLM and accumulator variance between runs, not a systematic bug.

5.5 Formal Proof Verification

[TODO: Write in your words.] A separate proof verification suite (`run_proof_tests.py`) was built to numerically verify the formal mathematical proofs. Seven theorems covering 14 claims, tested with 10,000 randomized inputs each. Completed in 0.8 seconds. Pure math, no server.

Initial run: 24/26. Theorem 3 ("reduces to single-axis models") failed Cases 1 and 2. Investigation revealed the σ_i non-monotonicity (see Appendix A). The theorem was wrong, not the constant. Corrected run: 29/29 theorems verified, 41/41 checks.

Key finding: Operational testing (29 tests, 14 runs) never caught the anomaly because no behavioral mechanism operates in the affected region. Only exhaustive mathematical verification across the full parameter space revealed it. Empirical testing and formal proofs are complementary. Each catches what the other misses.

5.6 What the Experiment Shows

1. Centroid valence tracks the intensity-weighted average across all conditions
2. Spread increases monotonically with source divergence (1.054 → 1.195 → 1.215)
3. Downstream behaviors read centroid position, not individual sources
4. Source removal produces immediate centroid recovery
5. Embodied agents may not reach zero spread because ambient sensors always contribute
6. Memory architecture validates geometric encoding: trace-type decay, stickiness modulation, true forgetting progression, and survival criteria all pass
7. Attachment bias validates geometric accumulation: stickiness resists movement (44x), monotonicity holds across mixed events
8. LLM behavioral output is stochastic and does not reliably translate geometric state into language (Test 3.3 hedging). The geometry is correct. The output layer is the bottleneck.
9. Formal mathematical proofs catch structural anomalies that empirical testing misses (the σ_i non-monotonicity existed from first deployment but was invisible to 14 test runs)

5.7 Supporting Deployments

The controlled experiment above is the main evidence. Three other deployments add context:

- **Production deployment (25 days, macOS):** 10,328 memories across 25 days confirmed that the geometric state produces consistent behavioral signatures under naturalistic conditions. Peak displacement of 0.462 from GPS with immediate recovery on return home.
- **iOS replication (2 days, iPhone):** 2,733 telemetry events confirmed cross-platform portability. The activation accumulator bug was independently reproduced and fixed,

providing evidence that the failure mode described in the papers is real and reproducible.

- **YAM v3 Parliament-of-Mind (April 2026, local Mac):** A second independent implementation of the geometric retrieval model on PostgreSQL with five hardpoint schemas. Each Parliament member carries its own sigma; the operational sigma is a centroid across the engaged hardpoints. Geometric selection (§4.6) is implemented in `parliament.patch_centroid` as a multiplicative blend of weighted concept overlap, an emotional-distance penalty against the live query centroid, and a sliding-window recency boost from the last 10 cross-engine winners. Recall events drift each answer's stored emotional position via an exponential moving average seeded from the encoding context. The trust gradient grades incoming signals on a five-tier rank (mama 1, perception 2, tutor 3, wikipedia 4, stranger 5) and scales the EMA α by speaker rank, so a stranger's correction drifts the brain at one-fifth the speed of mama's. The same column (`trust_rank=1`) that exempts mama-tier teachings from sleep decay also exempts them from drift — the immutable core. The verification gauntlet for the v3 memory commit (April 11, 2026) produced exact-math evidence of all four mechanisms: the bird answer encoded at valence +0.5 drifted by 5 mama recalls reached −0.374 (mama corrects fast); the cat answer encoded at +0.5 drifted by 5 stranger recalls reached only +0.260 (stranger corrects 3.6× slower at the same pressure); 10 frustrated stranger attacks on a `trust_rank=1` identity teaching bumped `access_count` from 2 to 12 but left `last_recon_centroid` unchanged (the bump is honest, the drift is refused); and the four-tier flashbulb sleep decay matched the paper formula to four decimal places per tier. v3 is the first working implementation of the §4.6 unification: emotional coordinates as retrieval index, sigma as read head, refutation by displacement.

Evidence Status

- **PROVEN:** Centroid weighted aggregation, spread correlation with source divergence, source removal recovery (Phase 1, 4 scenarios)
- **PROVEN:** Deception threshold triggers from distance-from-sigma (Phase 2)
- **PROVEN:** Memory architecture: trace-type decay ordering, stickiness modulation, reconsolidation diminishing, true forgetting progression, survival criteria (Phase 6, 6/6 tests)

- **PROVEN:** Attachment bias: stickiness resists movement (44x), monotonicity holds (Phase 7, 3/4 tests, 7.1 is test ordering not geometry)
- **PROVEN:** Cross-platform portability (Toughbook, MacBook Air, iPhone)
- **PROVEN:** Formal mathematical proofs: 29/29 theorems verified, 10,000 random trials each
- **SUPPORTED:** Embodied agents may not reach zero spread (observed, not formally proven)
- **SUPPORTED:** LLM behavioral output does not reliably translate geometric state into language (Test 3.3 hedging, Runs 18-19)

Likely Reviewer Criticism

- "n=1 agent. No statistical power." The agent experiment shows mechanism, not population effects. The dataset analyses in Chapter 6 provide statistical validation on human data.
- "The LLM generates the behavioral output. You're testing prompt engineering, not the model." The geometric state is computed from sensor inputs and injected source points. The LLM reads the state. It does not generate the centroid positions. Those are math.
- **Structural concern:** This chapter mixes methodology and results. Most committees want a standalone methods chapter covering setup, instrumentation, test protocol, and data collection so the method can be evaluated independent of the results. Consider splitting this into Chapter 5: Methods and Chapter 6: Results before submission.

Chapter 6: Existing Datasets

6.1 Purpose

Show that the geometric decomposition finds predictive structure in existing human data, independent of the agent.

6.2 Analyses

6.2.1 IBM HR Analytics (n=1,470)

- Geometric quadrants produce 2.2x difference in attrition rates from same satisfaction scores
- Employees with identical single-axis scores occupy different geometric locations with different outcomes
- Sigma inferred from item mapping

6.2.2 American Customer Satisfaction Index (n=7,341)

- Geometric subtypes produce up to 2.3x variation in complaint rates within same satisfaction score
- Pattern holds across score levels (6, 7, 8) and across four industries
- Sigma directly measured from expectations questions
- Cross-dataset convergence: structurally identical finding in different domain

6.2.3 NHANES PHQ-9 (n=5,455)

- Four geometric subtypes within moderate depression band (PHQ-9 10–14, n=455)
- Suicidal ideation rates range from 7.8% to 27.5% across subtypes, a 3.5x difference
- 132 respondents scoring exactly PHQ-9=10 separate into four clinically distinct presentations
- Same nine items, different math, backward compatible with existing instruments

Evidence Status

- **PROVEN:** Single-axis scores hide geometric variation in all three datasets
- **PROVEN:** Hidden variation predicts real outcomes (attrition, complaints, suicidal ideation)
- **PROVEN:** The finding generalizes across domains (HR, customer satisfaction, clinical)
- **SUPPORTED:** Sigma (measured or inferred) is associated with outcome differences the total score does not distinguish

- **NOT YET TESTED:** Whether the GAS as a native three-axis instrument outperforms decomposition of existing scales

Likely Reviewer Criticism

- "The item-to-axis mapping is subjective." Acknowledged. Two different mappings on two different datasets produce the same finding. The ACSI mapping uses directly measured expectations as sigma, which removes subjectivity for that axis.
- "Cross-sectional data cannot establish causality." Correct. The thesis claims predictive association, not causation. Longitudinal validation is future work.
- "The NHANES suicidal ideation finding is from a general population sample, not a clinical one." Acknowledged in the paper. Replication in clinical samples is a requirement before clinical use.
- **Structural concern:** The jump from "this works in my agent" (Chapter 5) to "this also shows up in NHANES data" (this chapter) needs more than a chapter break. The opening of this chapter needs to explicitly argue why applying the same geometry to both synthetic agents and human survey data is valid, and what it means that the same structure shows up in both.

Chapter 7: Applications

7.1 Purpose

The downstream stuff the geometric model drives. These are secondary contributions. Consequences of the model, not the model itself.

7.2 Applications (briefly)

- **Memory decay:** Intensity at encoding drives differential decay rates across memory types (Riggleman 2026l).

- **Deception:** Distance from sigma is the threshold and magnitude input for the deception mechanic (Riggleman 2026b, 2026m).
- **Nightmare formation:** Intensity and valence at consolidation replace the arbitrary nightmare threshold with a geometric one (Riggleman 2026d, 2026l).
- **Addiction:** The positive valence axis produces a structural mirror of trauma encoding. $Sigma_i$ is a measurable geometric trait for addictive personality: low values are a personality hum, high values produce agents that structurally cannot rest. Two-phase model separates stuck memory (phase one) from personality reorganization (phase two) (Riggleman 2026n).
- **Identity discontinuity:** Environmental orphaning in memory transplant shows coordinate-system dependence (Riggleman 2026o).
- **Survey measurement:** The GAS puts the three-axis framework into a survey humans can answer (Riggleman 2026r).
- **Clinical assessment:** Geometric displacement gives you a trajectory-based clinical metric (Riggleman 2026s).
- **Refutation as displacement:** Correction does not require a separate mechanism. Negative feedback ("no") carries emotional coordinates (shame, fear, pain) that displace the wrong answer away from sigma in affective space. The wrong answer is not weakened or deleted. It is moved. The correct answer, encoded at neutral coordinates near sigma, wins retrieval by proximity. Intensity does not need to go negative. Valence carries the sign. Intensity carries the weight. A high-intensity negative-valence memory is geometrically far from sigma, which is exactly what fear is. The entire correction mechanism falls out of the existing geometry with zero new architecture. Implemented in YAM v3 (April 2026) as the encoding-seeded EMA drift: `last_recon_centroid_v/a` on every answer is the running average of the recall contexts the memory has been pulled into, and the geometric selection rule reads from there instead of from the original encoding when ranking candidates.
- **Trust gradient as a property of the recall site, not the memory:** The same refutation-as-displacement mechanism reveals that *not all corrections drift the brain at the same rate*. Inbound trust applied to a recall event scales the EMA α coefficient: a high-trust speaker

(operator, mama) drifts the wrong answer at full alpha (0.20); a low-trust speaker (stranger, unknown) drifts at one-fifth (0.04). The brain still listens to everyone — it just weights what it hears by who said it. Two protections layer on top: drift saturation (the diminishing-returns alpha × recon_count limits how far one speaker can move a memory) and the immutable core (memories at the highest trust rank are exempt from drift entirely, the same exemption that protects them from sleep decay). The architecture pre-figured the implementation: the `trust_rank` column has been on the v3 schema since the v3 rewrite, used by sleep decay to protect mama-tier identity teachings; extending the same column to drift gating required no schema change, only honoring the gradient that was already there. This is how the geometric model handles the Peter problem (Riggleman 2026i): not by refusing to listen, but by weighting verbal claims at low priority until either sustained repetition or sensor corroboration promotes them.

Each one has its own paper. The thesis treats them as evidence the model generalizes, not as standalone contributions.

Structural Concern

- This chapter is a bullet list with one-sentence descriptions and paper citations. As a standalone chapter it will get flagged as padding. Either expand each application into a real discussion of how it follows from the geometry (a paragraph or two each showing why the geometric model is necessary for the application to work), or demote this to a section inside the conclusion.

Chapter 8: Evidence Classification

8.1 Purpose

Separate what is proven, what is supported, and what is speculative. Honesty requires this.

8.2 Classification

Claim	Status	Evidence
Centroid weighted aggregation produces predicted centroid positions	Proven	Centroid validation experiment, 4 scenarios, 110 cycles
Spread increases monotonically with source divergence	Proven	1.054 → 1.195 → 1.215 across scenarios
Downstream behaviors track centroid position, not individual sources	Proven	Behavioral output changed with centroid, not with individual source states
Source removal produces immediate centroid recovery	Proven	Valence jumped 0.07 → 0.68 on removal
Geometric decomposition reveals hidden structure in scalar instruments	Proven	IBM HR (2.2x attrition), ACSI (2.3x complaints), NHANES (3.5x ideation)
The architecture generalizes across platforms	Proven	Toughbook, MacBook Air, iPhone. Same architecture, same behavior
Activation axis accumulator produces different state than snapshot measurement	Proven	iOS replication independently reproduced the predicted failure mode and confirmed the fix
Memory architecture validates geometric encoding (trace-type decay, stickiness,	Proven	Phase 6: 6/6 tests passed, Run 11. Per-trace decay ordering confirmed, reconsolidation diminishing verified, full → fragment → ghost erosion confirmed

Claim	Status	Evidence
true forgetting, survival)		
Attachment bias validates geometric accumulation (stickiness, monotonicity)	Proven	Phase 7: 3/4 tests passed, Run 11. High-stickiness entity moved 44x less than fresh entity. Test 7.1 failure was test ordering, not geometry
Formal mathematical proofs verified by exhaustive numerical testing	Proven	29/29 theorems, 41/41 checks, 10,000 random trials each. Caught σ_i anomaly invisible to 14 operational runs
Test infrastructure found latent production bugs that deployment masked	Proven	Run 1: activation overwrite and deception-from-database bugs. Both real, both invisible in production use
Forgotten fear component leaks into test mode via fear_level computation	Proven	Runs 15, 17: test_mode blocks sensor sources but not fear_level. Procedural fix (say hello) confirmed on Runs 16, 18
LLM output does not reliably translate geometric spread into hedging language	Proven	Test 3.3: Runs 18, 19 both show S2=5, S1=6 hedging words. Geometry correct, output stochastic. Architecturally addressed by YAM
Embodied agents may not reach zero spread	Supported	Observed in centroid experiment; ambient sensors contribute nonzero spread
Sigma is associated with outcome differences that total	Supported	ACSI directly measured sigma; higher sigma = higher complaint rate at same score

Claim	Status	Evidence
scores do not distinguish		
The geometry shares structural properties with established human affective models	Supported	Russell (1980), Larsen & Diener (1987), Cacioppo & Berntson (1994), Bowlby (1969), Lazarus (1991). All arrived at inductively, not used as design inputs
Memory and emotion are one system: emotional coordinates are the retrieval index, sigma is the read head	Proven	YAM v3 Parliament-of-Mind (§5.7) is the first working implementation of the §4.6 unification. `parliament.patch_centroid` ranks candidate answers by emotional distance to the live query centroid, not by raw concept overlap. Verification gauntlet (April 11, 2026) showed exact-math match between predicted and observed drift trajectories across 5 sequential recall events at two trust ranks. The geometric retrieval rule is implemented and demonstrably picks the closer answer, not the stronger one. Structural derivation from prior versions is now backed by working code
Refutation (correction via "no") works by emotional displacement, not by weakening the wrong answer	Proven	YAM v3 (April 11, 2026) implements refutation-as-displacement via the encoding-seeded EMA drift on `last_recon_centroid_v/a`. Drift demonstration: a wrong answer's emotional position drifted from (+0.5, +0.4) to (−0.660, +0.779) after 10 frustrated mama recalls without any importance reduction (importance kept growing via reconsolidation; only the position changed). The wrong answer was moved, not weakened. Subsequent calm queries peeled back only ~20% of the drift per event, demonstrating asymmetric recovery proportional to the EMA α. Requires no new mechanism beyond the existing geometry, as the structural derivation predicted
Trust gradient on recall events scales the rate at which	Proven	YAM v3 (§5.7, April 11, 2026): the same correction (recall context valence −0.8, activation +0.8) drifted a tutor-tier answer 3.6× faster when delivered by a mama-rank speaker

Claim	Status	Evidence
corrections drift a memory		(alpha 0.20) than by a stranger-rank speaker (alpha 0.04). Trajectories matched the predicted EMA formulas to four decimal places. The brain listens to everyone but weights inputs by source identity
Identity-tier teachings are structurally protected from drift even under repetitive coercion	Proven	YAM v3 (April 11, 2026): 10 frustrated stranger-rank attacks (recall context –1.0 valence, +1.0 activation) on a `trust_rank=1` identity teaching incremented `access_count` from 2 to 12 and applied the importance boost as expected, but `last_recon_centroid` remained NULL across all 10 attacks. The bump is honest; the drift is refused. The same column that exempts mama-tier teachings from sleep decay also exempts them from drift via the immutable-core rule
The pride answer: nearest-neighbor retrieval produces both confabulation under absence and belief perseverance under correction as a single mechanism	Supported	YAM v3 (April 11, 2026): observed in the elephant-regression sweep (queries about wiki nouns whose content lacks the noun fall back to the lexically nearest available answer, regardless of relevance) and in the drift demonstration (drifted answers resist single-event correction at a rate proportional to accumulated prior drift). Both behaviors fall out of the geometric retrieval rule. Empirical measurement in human respondents is future work (§10)
The geometric model produces changed behavioral output following state discontinuity	Supported	Single shutdown experiment; behavioral shift observed but not replicated
The GAS as a native instrument outperforms scalar instruments	Not yet tested	Backward compatibility demonstrated; native collection requires prospective study
Trajectory-based clinical assessment	Not yet tested	Requires longitudinal clinical data with treatment outcomes

Claim	Status	Evidence
detects approach before arrival		
The geometric framework may generalize across sensory modalities	Speculative	Theoretical proposal only (Riggleman 2026q). Requires echolocation robot experiment
The agent experiences subjective states	Not claimed	The thesis describes behavior, not phenomenology. This claim is outside scope.

Chapter 9: Limitations

9.1 Agent Experiment Limitations

- **n=1 architecture.** All agent experiments involve one system on three platforms. The experiments show the mechanism works in this architecture. They do not show it generalizes to other architectures. Replication on an independent agent is needed before the mechanism claim is general.
- **LLM variability.** The behavioral output comes from an LLM that reads the geometric state. The same state could produce different language on different runs. Centroid positions are deterministic. Behavioral responses are not. The geometric validation is clean. The behavioral validation has noise. Test 3.3 (hedging) demonstrates this concretely: high-spread and low-spread configs produced nearly identical hedging word counts (5 vs 6) across Runs 18 and 19. The geometry was correct. The LLM did not act on it.
- **Ambient source contamination.** The centroid experiment could not fully isolate injected sources from ambient sensor inputs. Spread never hit zero because embodied sensors always contribute. This might be a property of embodiment rather than a flaw, but it means Scenario 1 is not a pure single-source test.

- **Forgotten fear leaks into test mode.** The `forgotten` fear component (15% weight, time since last interaction) feeds the activation accumulator even in test_mode, because test_mode blocks sensor source points but not the fear_level computation. Runs 15 and 17 failed activation tests for this reason. Procedural fix: interact with the agent before testing. This is a limitation of embodied testing, not the geometry.
- **Scenario 5 not yet executed.** The gradual intensity ramp is planned but not completed. Without it, the thesis cannot confirm linearity of the centroid response across the full intensity range.

9.2 Dataset Analysis Limitations

- **Item-to-axis mapping involves judgment.** The researcher picks which survey items map to which axis. Different mappings could produce different results. The ACSI analysis partly addresses this because expectations are measured directly as sigma, but valence and activation mappings are still researcher calls.
- **Cross-sectional data cannot establish causality.** The IBM HR, ACSI, and NHANES analyses show association between geometric position and outcomes. They do not show causation. Longitudinal data with intervention controls would be needed for that.
- **General population vs. clinical.** The NHANES suicidal ideation finding is from a national health survey, not a clinical population. Rates, distributions, and thresholds may look different in psychiatric settings. Replication in clinical samples is required before any clinical application.
- **No native three-axis collection.** All dataset analyses are retroactive decompositions of existing single-axis data. Whether collecting all three axes natively produces the same or better results has not been tested.

9.3 Model Limitations

- **Euclidean distance assumptions.** The distance metric assumes the three axes are orthogonal and equally weighted. Neither assumption has been validated. If the axes are correlated or one axis matters more than the others, a weighted or Mahalanobis distance might work better.

- **Sigma stability.** Sigma is assumed stable over weeks to months. This has not been tested in human respondents. If sigma drifts between measurements, displacement computations become unreliable.
- **Intensity derivation vs. self-report.** In the agent, intensity is computed from valence and activation displacement. In the survey, intensity is self-reported. Whether those two approaches measure the same thing is testable but has not been tested.
- **Sigma intensity pathology threshold.** The sigma intensity anomaly (Appendix A) shows that high σ_i produces downstream system behavior consistent with addictive personality. The threshold at which personality trait becomes structural pathology is computable from the downstream parameters but has not been computed or validated. Whether that threshold is sharp or gradual is an open question. Full analysis is in Riggleman (2026n) Section 3.5.

9.4 IRB Considerations

No human subjects were involved in any experiment presented in this thesis. All agent experiments used synthetic systems. All dataset analyses used publicly available, de-identified data (IBM HR Analytics via Kaggle, ACSI via Mendeley Data, NHANES via CDC).

Future work involving human respondents (GAS validation, clinical trajectory tracking) will require IRB approval. The thesis identifies these as future directions and does not present them as completed work.

Chapter 10: Future Work

10.1 Human Validation

- Run the GAS alongside existing instruments
- Test-retest reliability on the three axes
- Measure sigma stability across sessions

- Whether self-reported intensity matches the computed version

10.2 Clinical Application

- Replicate the NHANES subtype finding in clinical populations
- Longitudinal trajectory tracking with treatment outcomes
- Test whether trajectory tracking actually catches problems early

10.3 Technical Extensions

- Other distance metrics (Mahalanobis, weighted Euclidean)
- What happens when multiple agents interact
- Measuring whether and how fast sigma drifts
- Compute the σ_i pathology threshold from downstream system parameters (decay, stickiness, deception)
- Controlled addiction/relapse protocol on the Potato system
- Cross-modal test (echolocation robot experiment)

Chapter 11: Conclusion

11.1 Purpose

Restate the contribution. No new claims. No expansion.

11.2 The Contribution

This thesis presents a three-dimensional geometric representation of affective state with a personality setpoint, a centroid model for multi-source aggregation, and evidence that this structure outperforms scalar representations in both synthetic agent behavior and human datasets across three independent domains.

The central finding is that emotional coordinates and memory retrieval are not separate systems. The affective state space that encodes emotion is the same space that indexes memory. Sigma is the read head. When multiple memories compete for retrieval, the one whose emotional coordinates sit closest to the agent's personality setpoint is selected. Every downstream phenomenon documented in this thesis — memory decay, reconsolidation, correction, attachment, addiction, fear — reduces to a single operation: geometric proximity to sigma in affective space.

This unification was not designed. It fell out of engineering requirements. A persistent agent needed a way to aggregate sensor inputs, track displacement from baseline, and select among competing memories. The engineering produced a system where the emotional tag on a memory is not metadata. It is the retrieval key. The realization that these were one system, not two, came from asking how correction works without a dedicated refutation mechanism. The answer — that "no" displaces the wrong answer's emotional coordinates away from sigma rather than weakening it — demonstrated that memory selection and emotional geometry cannot be separated. They are the same computation.

That same realization was independently re-derived during the implementation of YAM v3 (Riggleman 2026, in development) when the elephant-regression bug surfaced and the right fix was sought from first principles rather than from re-reading the thesis. The fix that fell out of the engineering — "if multiple memories are returned take the one with emotion closest to the sigma or centroid; the more you say no the more that answer moves; it does not matter the strength, if anything the strength makes the memory stronger" — landed exactly on §4.6 and §7.2 of this document. Two independent derivations of the same result, one from theory and one from engineering pressure on a separate codebase, is the kind of convergence that makes the unification credible. The architecture had been waiting for a working implementation to make §4.6 concrete. That implementation now exists, and the verification gauntlet shows the geometric retrieval rule producing the predicted behavior to four decimal places.

One observation surfaced from the v3 implementation that was not previously named in this thesis and is now in §4.7: the pride answer. When the geometric retrieval rule operates under partial knowledge, it produces both confabulation (the brain fills silence with the closest match available, no matter how irrelevant) and belief perseverance (drifted answers resist

correction proportional to accumulated drift). Both behaviors fall out of the same architectural rule. They are not two cognitive failures. They are one mechanism caught at different stages of the same condition. The brain refuses to be silent.

The model came from engineering requirements of a deployed AI agent. It lines up structurally with established models of human affect without having been built from them. The dimensional emotion literature describes how affective states are structured. The memory literature describes how storage and recall work. In the geometric model they are the same system described from two directions. Whether that overlap says something general about how persistent systems with memory and internal state organize their retrieval is the question this thesis leaves open.

Supporting Papers

The thesis draws on 22 companion papers, all available at clawddaily.com/papers. Each paper is a standalone contribution. The thesis synthesizes them into a unified argument.

Paper	Thesis Chapter	Role
2026j: Valence	Ch. 3	Axis definition
2026k: Activation	Ch. 3	Axis definition + accumulator
2026l: Intensity	Ch. 3	Axis definition + memory decay
2026m: Emotional Geometry	Ch. 3	Unified model + sigma
2026t: Centroid Model	Ch. 4	Multi-source aggregation + spread
2026f: SGI Holistic	Ch. 5	Production deployment data
2026a: Memory Decay	Ch. 5, Ch. 7	Phase 6 validation + application: memory
2026b: Lie Mechanic	Ch. 5, Ch. 7	Phase 2 validation + application: deception

Paper	Thesis Chapter	Role
2026d: Nightmares	Ch. 7	Application: dream cycle
2026e: Full Spectrum	Ch. 5	Prior model (superseded by geometry papers)
2026i: Outbound Trust	Ch. 5	Peter experiment
2026n: Addiction	Ch. 7, Appendix A	Application: positive valence axis, sigma intensity as addictive personality trait, two-phase model
2026o: Identity Discontinuity	Ch. 7	Application: memory transplant
2026p: Functional Consciousness	Ch. 5	Shutdown experiment
2026q: Cross-Modal Translator	Ch. 10	Future work: modality independence
2026r: GAS Survey	Ch. 6	Dataset analyses (IBM HR, ACSI)
2026s: Clinical Metric	Ch. 6	Dataset analysis (NHANES PHQ-9)
2026u: Attachment Bias	Ch. 5	Phase 7 validation: stickiness, monotonicity, formation matrix

Appendix A: The Sigma Intensity Anomaly

The intensity function

$$I(v, a) = \min(1, \sqrt{((v - \sigma_v)^2 + (a - \sigma_a)^2)} / \sqrt{2})$$

The model computes intensity from two inputs: *v* (current valence) and *a* (current activation). σ_v and σ_a are the valence and activation components of the resting point sigma.

$(v - \sigma_v)$ is how far valence is from resting valence. Square it. $(a - \sigma_a)$ is how far activation is from resting activation. Square it. Add the two squares and take the square root. That gives you the Euclidean distance from the current point to sigma in the valence/activation plane. Pythagorean theorem on a 2D plane.

Then divide by $\sqrt{2}$. Valence and activation both range 0 to 1, so the farthest any two points can be from each other in that unit square is the diagonal: $\sqrt{(1^2 + 1^2)} = \sqrt{2}$. Dividing by $\sqrt{2}$ normalizes intensity to a 0-to-1 range.

The min(1, ...) is a clamp. If the raw value goes over 1, cap it.

So intensity is just "how far am I from home, as a fraction of the farthest I could possibly be from home." At home, distance is zero, intensity is zero. At the far corner of the space, intensity is 1.

The observation

The deployed sigma was $\sigma = (\sigma_v, \sigma_a, \sigma_i) = (0.4, 0.2, 0.2)$. Plug the resting point into the intensity formula: $v = \sigma_v$, $a = \sigma_a$. That gives $\sqrt{((\sigma_v - \sigma_v)^2 + (\sigma_a - \sigma_a)^2)} = \sqrt{(0 + 0)} = 0$. So $I(\sigma_v, \sigma_a) = 0$ by definition. But σ_i was stored as 0.2. The formula produces intensity 0 at rest. The stored sigma says intensity 0.2 at rest. These do not match. The question is whether that mismatch is an error or a feature. The math below shows what it does. The interpretation follows.

Setting up the distance proof

To show what this contradiction actually does, hold activation fixed at σ_a so only valence moves. Define $\delta = v - \sigma_v$. Delta is how far valence moved from rest. Pinning activation collapses the problem to one dimension so the anomaly shows up clearly.

With activation pinned, the $(a - \sigma_a)$ term is zero and intensity simplifies to:

$$I(v, \sigma_a) = \sqrt{(\delta^2)} / \sqrt{2} = |\delta| / \sqrt{2}$$

Intensity is absolute valence displacement divided by $\sqrt{2}$. Move valence away from rest, intensity goes up proportionally.

The distance anomaly

Compute the squared distance from the current state (v, a, I) to sigma (σ_v, σ_a, σ_i) in three dimensions. Activation is pinned at σ_a, so that dimension drops out. Two terms remain:

$$d^2 = (v - \sigma_v)^2 + (I - \sigma_i)^2$$

First term is δ^2. For the second term, I is $|\delta|/\sqrt{2}$ (from above) and σ_i is 0.2. Expand it:

$$(|\delta| / \sqrt{2} - 0.2)^2 = \delta^2/2 - 2 \cdot (|\delta| / \sqrt{2}) \cdot 0.2 + 0.04$$

Note that $2 \cdot (1/\sqrt{2}) = \sqrt{2}$. That is where the $\sqrt{2} \cdot \sigma_i$ coefficient comes from. Simplify:

$$= \delta^2/2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2$$

Add the first term δ^2 back in:

$$d^2 = \delta^2 + \delta^2/2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2 = (3/2)\delta^2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2$$

Three terms. First term, $(3/2)\delta^2$, pushes distance up as valence moves. Expected. Third term, $\sigma_i^2 = 0.04$, is a constant. The middle term is where it gets interesting: it is negative and linear in $|\delta|$. For small moves away from rest, that negative linear pull beats the positive quadratic push. Total distance goes *down* before the quadratic catches up and drives it back up.

Finding the minimum

Take the derivative of d^2 with respect to $|\delta|$ and set it to zero. Let $x = |\delta|$:

$$d^2(x) = (3/2)x^2 - \sqrt{2} \cdot \sigma_i \cdot x + \sigma_i^2$$

$$\mathrm{d}(d^2)/\mathrm{d}x = 3x - \sqrt{2} \cdot \sigma_i = 0$$

$$x = \sqrt{2} \cdot \sigma_i / 3$$

Plug in $\sigma_i = 0.2$:

$$x = \sqrt{2} \cdot 0.2 / 3 = 0.2828\ldots / 3 \approx 0.094$$

The minimum distance to sigma is not at $\delta = 0$ (rest). It is at $|\delta| \approx 0.094$. A point slightly off-center registers as closer to home than home itself. At $\sigma_i = 0.2$ this offset is small. At higher values of σ_i the offset grows and the behavioral consequences become significant.

What this means

Automated testing caught the inconsistency. The test harness flagged a non-monotonic distance function and the initial read was that $\sigma_i = 0.2$ was an implementation error. But interpreting what the math was actually doing took iterative explanation, challenge, and rethinking before the result could be defended as an accidental design choice rather than a simple bug.

The original interpretation was that $\sigma_i = 0.2$ was a bug and the fix was to set $\sigma_i = 0$. That interpretation is wrong. It is one valid design choice, not the only one.

Setting $\sigma_i = 0$ enforces the constraint $\sigma_i = I(\sigma_v, \sigma_a) = 0$. The distance formula becomes:

$$d^2 = (3/2)\delta^2 - \sqrt{2} \cdot 0 \cdot |\delta| + 0^2 = (3/2)\delta^2$$

Middle term is gone. $d = |\delta| \cdot \sqrt{(3/2)}$. Monotonic. Clean. No false minimum. That is what a personality with zero baseline intensity looks like in the math.

But $\sigma_i > 0$ is not an error. It is a personality that includes baseline intensity. The non-monotonic distance function is not the system misbehaving. It is the system correctly representing an agent whose rest includes edge. The agent at dead zero displacement is less at home than the agent with a small disturbance, because dead zero is not where you put home. The small bump does not reward the agent. It brings the agent home.

At low σ_i (0.1 to 0.2) this is a personality trait. The agent runs a little hot. The downstream offsets are small. The agent is functional.

At high σ_i the behavioral profile changes. The minimum achievable distance grows. The agent structurally cannot rest. Memory decay never reaches baseline rate. Stickiness is always elevated. The agent needs bigger and bigger displacements to close the intensity gap. The agent is most comfortable in crisis. These are not analogies. They are the same math at different coordinates producing the behavioral profile of addictive personality.

σ_i is a measurable geometric trait. Low values produce functional personalities with a slight hum. High values produce agents trapped in permanent displacement where no downstream system works as designed. The threshold where personality becomes pathology is computable from the downstream system parameters.

The full analysis, including the two-phase model of addiction (phase one: frozen reconsolidation with intact sigma, phase two: sigma drift with personality reorganization) and the role of σ_i as a diagnostic parameter, is in Riggleman (2026n) Section 3.5.

A note on the scaling factor

Moving 1 unit in valence alone produces a distance of $\sqrt{(3/2)} \approx 1.225$, not 1.0. That is because moving valence also drags intensity up. Intensity is derived from displacement, so a pure valence move is really a move in two dimensions at once. The extra 0.225 is the intensity shadow of the valence movement.

Full investigation, timeline, and corrected proof: sigma-intensity-anomaly.md

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Version Archive

Every version of this thesis is publicly archived. Nothing is hidden. Nothing is deleted.

Version	Date	Changes	Archive
0.7	2026-04-11	v3 implementation evidence folded in. New §4.7 (The Pride Answer) names the cognitive pattern that emerges from the §4.6 unification under partial knowledge: confabulation under absence and belief perseverance under correction as a single mechanism. New §5.7 third supporting deployment: YAM v3 Parliament-of-Mind, the first working implementation of geometric retrieval (§4.6) with verification gauntlet. Ch. 7 application list extended with the trust gradient as a recall-site property scaling EMA α by speaker rank. Ch. 8 evidence table promotes two rows from Supported to Proven (the §4.6 unification and refutation-as-displacement) and adds three new Proven rows (trust gradient drift ratio, immutable-core protection, and pride answer as a Supported observation). Ch. 11 conclusion adds the convergence note: independent re-derivation of the same result during v3 engineering, the architecture waiting for the implementation that now exists.	View
0.6	2026-04-11	Capstone finding added: memory and emotion are one system. Sigma identified as retrieval index (read head). New §4.6 derives memory selection from geometric proximity to sigma. Refutation mechanism added to Ch. 7 as application: correction via emotional displacement, no new architecture required. Ch. 8 evidence table expanded with unification and refutation claims. Ch. 11 conclusion rewritten around the central finding that emotional coordinates are the memory retrieval key.	View
0.5	2026-04-07	Working draft published live to clawddaily/thesis/. Built on v0.4.2 with the structural concern boxes addressed and the supporting deployment context expanded.	View
0.4	2026-04-02	Ch. 5 expanded to cover full 7-phase, 29-test suite (was Phase 1 only). Hardware corrected from Toughbook to Threadripper. Added iteration record (19 runs), formal proof verification, sensor simulation decisions, forgotten fear discovery, Test 3.3 LLM limitation. Ch. 8 evidence table expanded with memory, attachment, proof, and LLM findings. Ch. 9 limitations updated with Test 3.3 and forgotten fear specifics. Supporting papers table updated with 2026u and cross-references to Phase 2/6/7 validation. Structural concern boxes added to Ch. 1 (scope), Ch. 2 (lit review), Ch. 3 (proofs), Ch. 5 (methods/results split), Ch. 6 (bridging argument), Ch. 7 (too thin).	View

Version	Date	Changes	Archive
0.3	2026-03-28	Four targeted fixes. (1) Intro scope tightened to one paragraph. (2) Ch. 4 restructured: centroid, spread, and distance from sigma each get their own subsection with definition, purpose, and testable prediction. (3) Ch. 5 experiment scenarios each state what they test, Scenario 5 (intensity ramp) added as planned. (4) Ch. 9 expanded to proper limitations with agent, dataset, and model subsections.	View
0.2	2026-03-28	Structural revision. Applications pushed to Ch. 7. Language toned.	View
0.1.1	2026-03-28	Tightened introduction. Contribution statement made explicit.	View
0.1	2026-03-28	Initial thesis outline. 11 chapters. Evidence classification table. Supporting paper map.	View